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Course: Calculus Beta

Assignment: MyLab Math - Uge 8

For mutually exclusive events R_1 , R_2 , and R_3 , we have $P(R_1) = 0.06$, $P(R_2) = 0.3$, and $P(R_3) = 0.64$. Also, $P(Q | R_1) = 0.2$, $P(Q | R_2) = 0.5$, and $P(Q | R_3) = 0.4$.

Find $P(R_1 | Q)$.

Use Bayes' Theorem to calculate your answer for this problem. Using the variables given for this problem, the formula may be expressed as follows.

$$P(R_1 | Q) = \frac{P(R_1) \cdot P(Q | R_1)}{P(R_1) \cdot P(Q | R_1) + P(R_2) \cdot P(Q | R_2) + P(R_3) \cdot P(Q | R_3)}$$

First, use the given values to calculate the numerator.

$$P(R_1) \cdot P(Q | R_1) = 0.012$$

Calculate the denominator.

$$\begin{aligned} P(R_1) \cdot P(Q | R_1) + P(R_2) \cdot P(Q | R_2) + P(R_3) \cdot P(Q | R_3) \\ = 0.06 \cdot 0.2 + 0.3 \cdot 0.5 + 0.64 \cdot 0.4 \\ = 0.012 + 0.15 + 0.256 \\ = 0.418 \end{aligned}$$

Write the result as a fraction in lowest terms.

$$\frac{0.012}{0.418} = \frac{6}{209}$$

$$\text{Therefore, } P(R_1 | Q) = \frac{6}{209}.$$